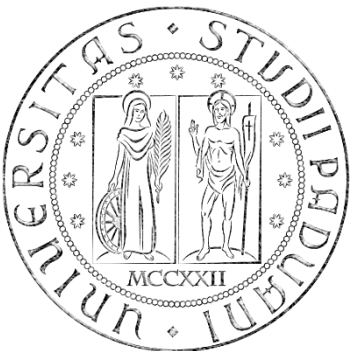


Biomedical Wearable Technologies for Healthcare and Wellbeing

Wearables

A.Y. 2025-2026
Giacomo Cappon

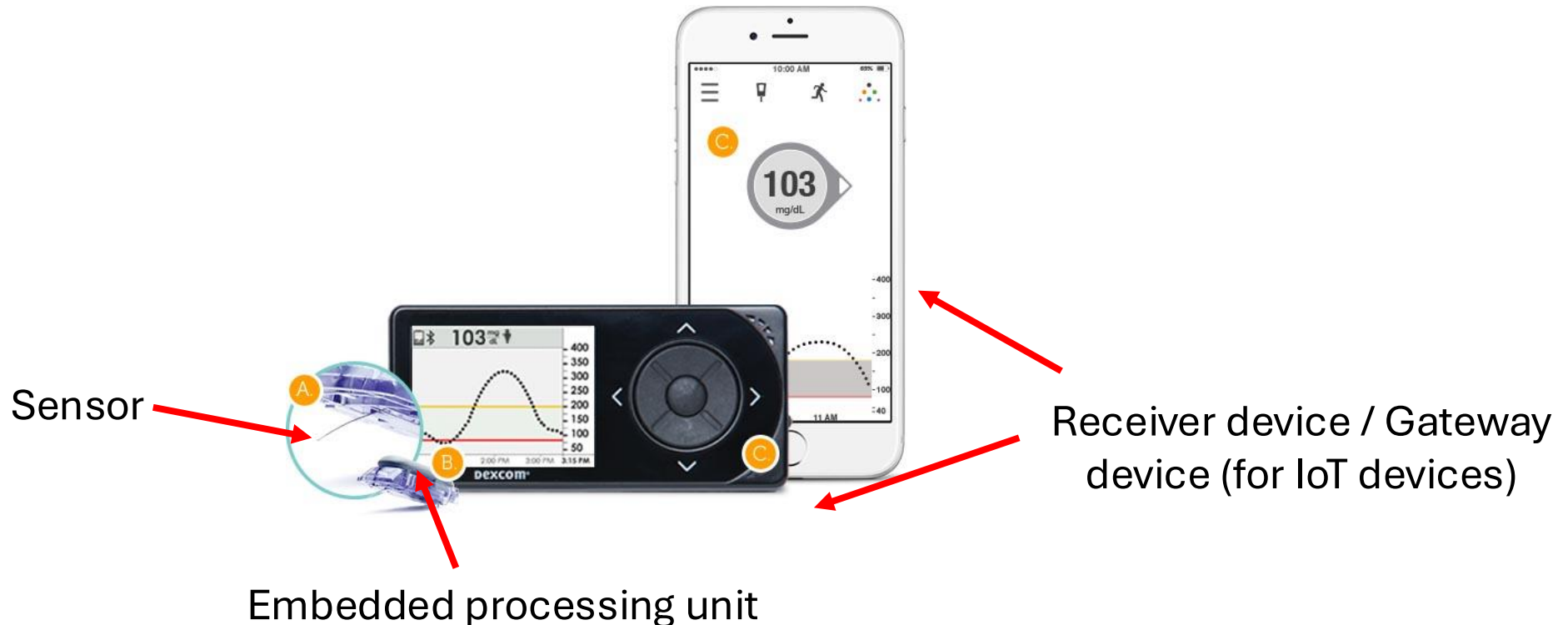


Wearable devices



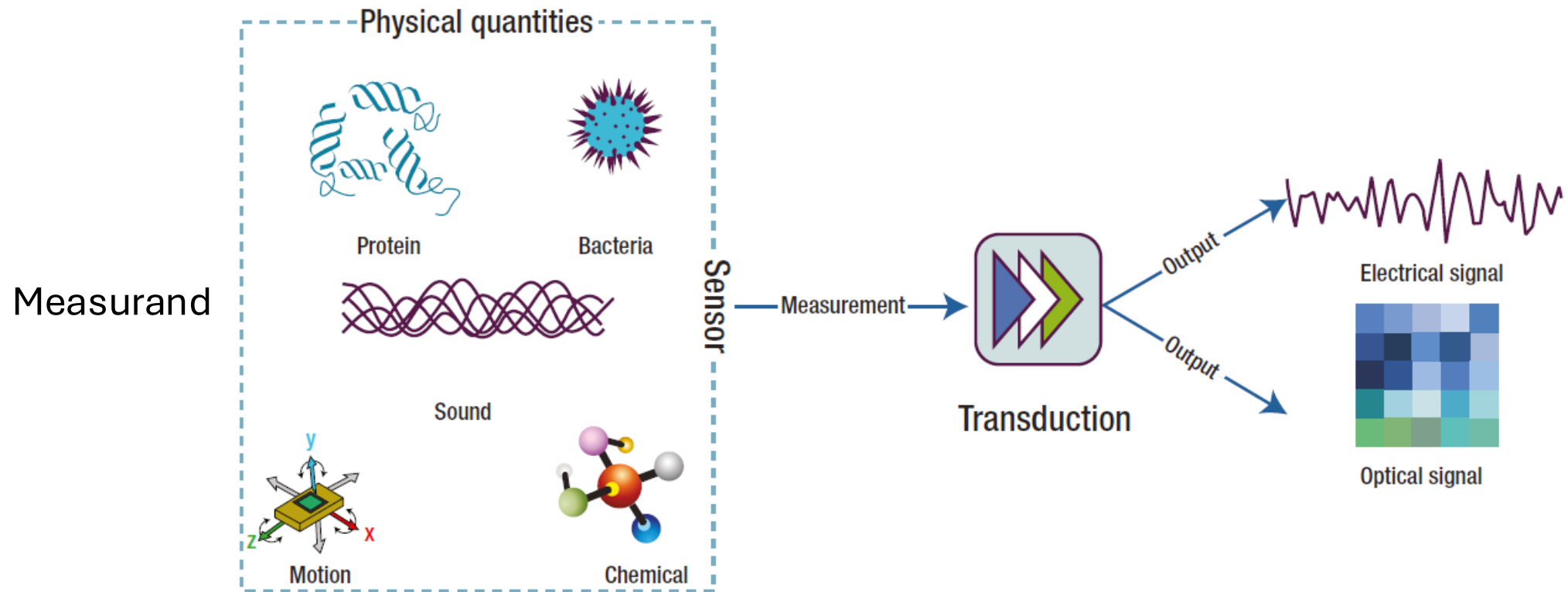
Wearable devices

- A wearable device is a device that can be used for sensing physiological parameters or signals which generally include two components:
- A **sensor** that measures the physical quantity of interest (measurand)
 - An **embedded processing unit** that does some initial processing of the measurements and either stores them in a memory or sends them to another device



Definitions of sensor

- “A sensor generally refers to a device that converts a physical measure into a signal that is read by an observer or by an instrument” (Chen et al., 2012)

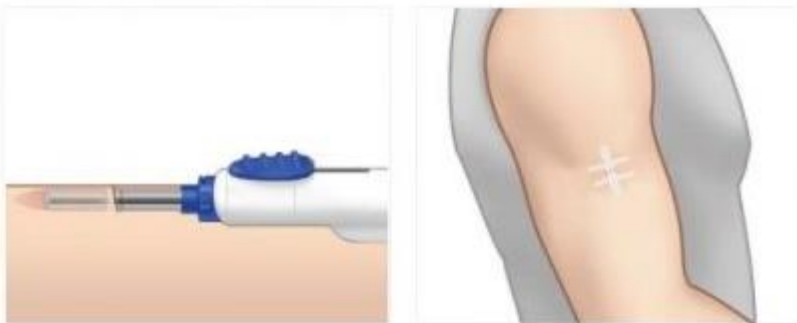


Classification of sensors based on their interaction with the measurand

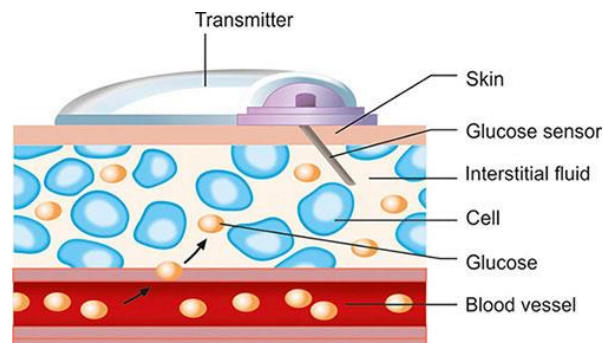
Contact sensor: sensors requiring physical contact with the measurand.

- Their deployment obviously perturbs the state of the subject to some degree.
 - Comfort and biocompatibility are important considerations
 - Response time determined by the speed at which the measurand is transported to the measurement site.
 - Poor contact can result in signal noise and the introduction of signal artifacts.

Invasive: sensors inserted into human body



Minimally-invasive: patch-type devices on the skin that monitor interstitial fluids



Non-invasive: sensors that simply have contact with the skin without effect



Classification of sensors based on their interaction with the measurand

Non-contact sensors: do not require direct contact with the measurand.

- Minimum perturbation of the subject
- Generally used in ambient sensing applications
- Example: infrared thermometer

Sample based sensors: require an invasive collection of a sample of the person to be monitored by a human or automated sampling system.

- The collected samples are analyzed using sensors or laboratory-based analytical instrumentation
- Common approach in healthcare (e.g. monitoring cholesterol)

Classification of sensors based on their sensing technique

- Mechanical sensors
- Electrical sensors
- Optical sensors
- Chemical sensors

Mechanical sensors

➤ **Mechanical sensors** detect some form of mechanical deformation of a material, in response to the measurand (input), and translate it to an electrical, optical, magnetic or thermal signal (output).

➤ **Electromechanical sensors:**

Measurand → Mechanical deformation → Electrical signal

➤ Types of electromechanical sensing techniques:

- Piezoresistive sensors
- Capacitive sensors
- Iontronic sensors
- Piezoelectric sensors

Electrical sensors

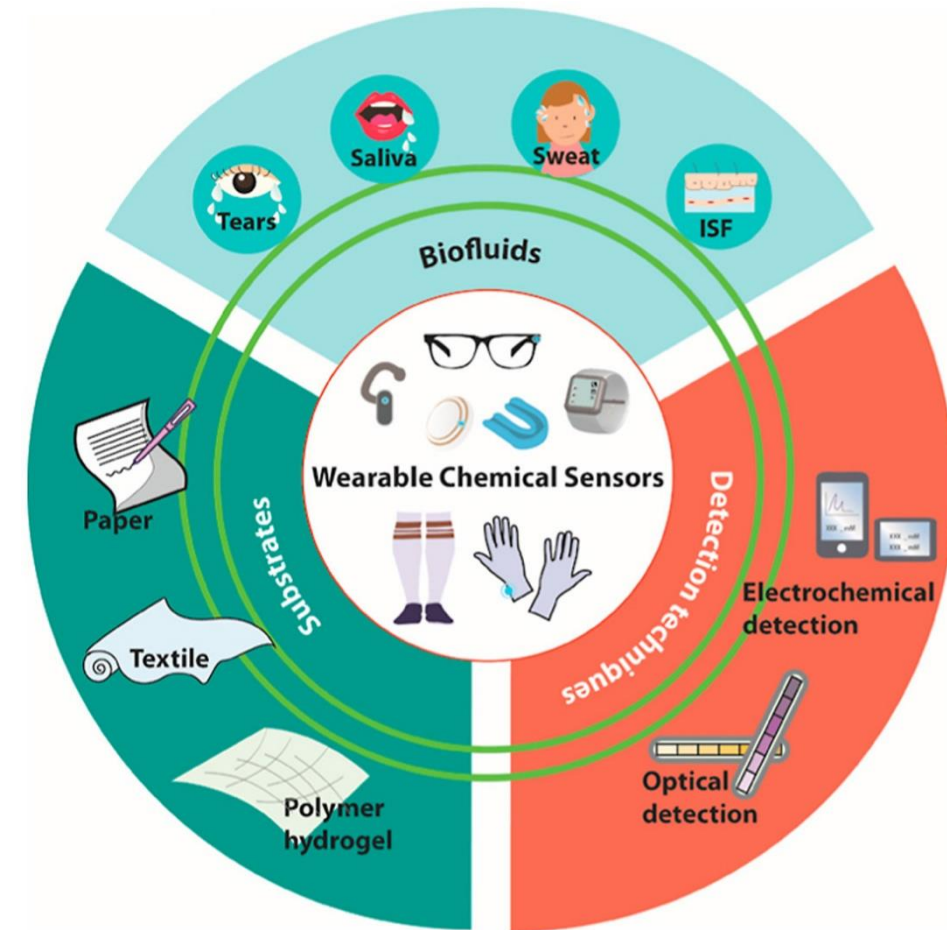
- **Electrical sensors:** on-skin electrodes that measure a change in electrical resistance or the electrical charge of the skin.
- Good electrical contact with skin is required.
- **Wet electrodes:** solid conductive pad interfaced to the skin via an electrolyte gel that minimizes the impedance of skin by hydrating it and forming a conformal electrical contact with its textured surface.
- **Dry electrodes:** direct contact with the skin, sensible to movement artifacts.
- Skin electrodes allow to collect high-fidelity measurements of a broad range of physiologically relevant biopotentials:
 - electrocardiogram (ECG)
 - electroencephalogram (EEG)
 - electromyogram (EMG)
 - electrooculogram (EOG)
 - electroretinogram (ERG)
 - galvanic skin response (GSR), also known as skin impedance or electrodermal activity (EDA)



Optical sensors

- **Electro-optical sensors** detect light and convert it to an electrical signal
 - *Photoconductive* devices convert a change of incident light into a change of resistance.
 - *Photovoltaics* (solar cells) convert an amount of incident light into an output voltage.
 - *Photodiodes* and *phototransistor* convert an amount of incident light into an output current.
- Operating principle of wearable optical sensors: a light source introduces light into the body and based on how much the light is absorbed or reflected, measurements related to the body tissue are derived.
- The wavelength of light sources can range from UV to deep infrared, depending on needed penetration depth and the absorption peak for the relevant sensing application.
- Commonly used for monitoring of heart rate and blood oxygenation.

Chemical sensors



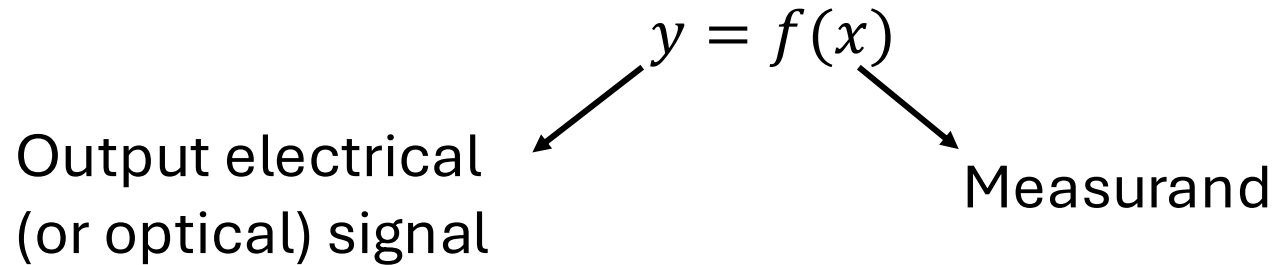
- They quantify chemical markers, or analytes, in biological fluids and convert this information in an optical (**optochemical sensor**) or electrical signal (**electrochemical sensor**).
- Require direct chemical interaction with the biomarker (contact sensors).
- Wearable chemical sensors use non-invasive or minimally-invasive techniques to measure analytes in **alternative biofluids** instead of invasive blood sampling.
 - Biofluids secreted by the body (saliva, sweat, tears)
 - Interstitial fluid reached through minimally-invasive sensors or by reverse iontophoresis
- When a new wearable chemical sensor is designed, the relationship between analyte concentration in the chosen biofluid and that in the traditional samples (e.g., blood) must be confirmed.

Sensor characteristics

- To truly understand sensors, and how sensors that measure the same measurand can differ, it is necessary to understand sensor performance characteristics, which are reported in **specifications** (specs) or **datasheet**.
 - Systematic characteristics: those which can be exactly quantified by mathematical or graphical means;
 - Statistical characteristics: those which cannot be exactly quantified;
 - Dynamic characteristics: those characterizing the ways in which an element responds to sudden input changes.

Transfer function

- A function f that describes the relationship between the measurand and the output signal (electrical or optical).

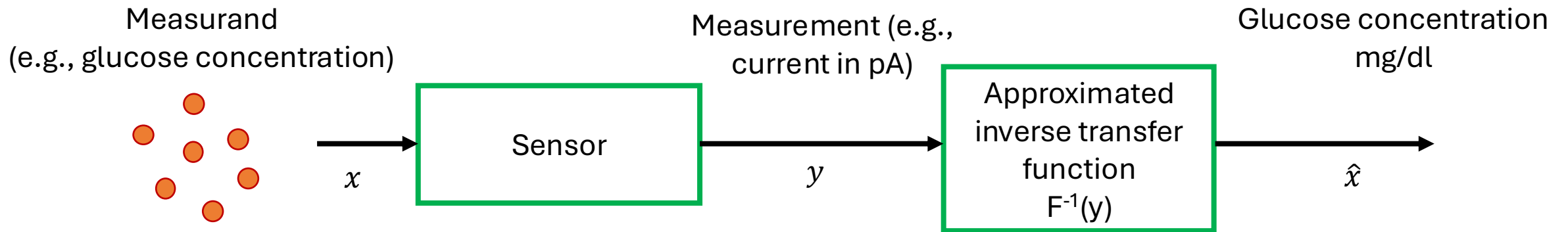


- Represented in specs as a table of values, a graph, or a mathematical formula.
- Typically the actual transfer function is complex → the transfer function is approximated by a simpler linear or non-linear function, $F(.)$, which can be mathematically expressed as a formula.

$$y = F(x) + e \quad e = \text{approximation error}$$

Inverse transfer function

- The embedded processing unit of the sensor device applies the inverse of $F(\cdot)$, $F^{-1}(\cdot)$, to convert the electrical signal generated by the sensing technique to output measurements in the same domain of the measurand.



Common transfer function approximations

Linear transfer function approximation:

$$y = a \cdot x + b + e$$

- The offset, b , is the output value of the sensor when no measurand is applied. It is the component of the output signal which does not depend on the measurand.
- The slope, a , represents the **sensor sensitivity**, i.e., the change in the sensor output resulting from a unit change in the input measurand.

Non-linear transfer function approximations:

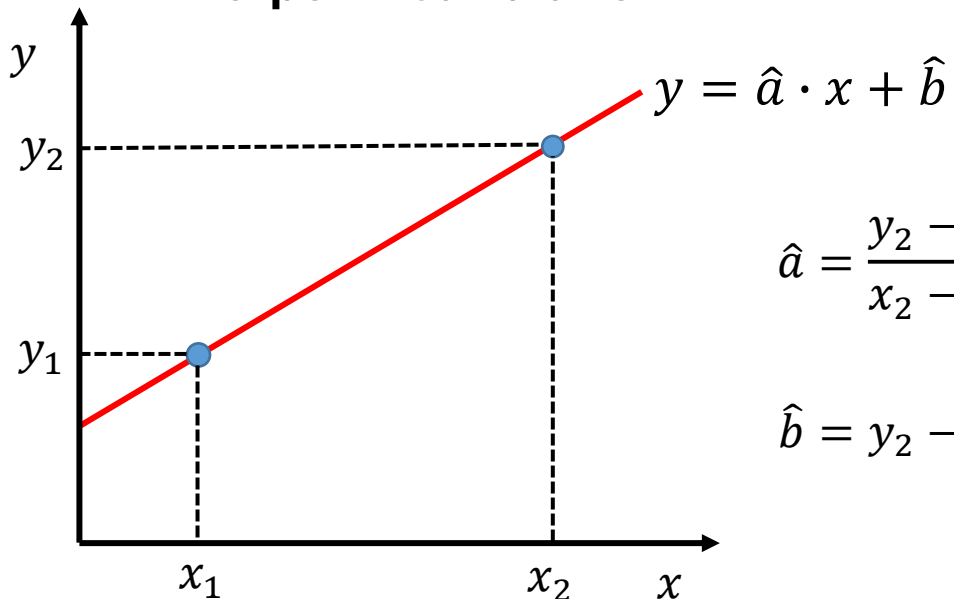
Logarithmic function	$y = a + b \cdot \ln(x) + e$
Exponential function	$y = a \cdot e^{k \cdot x} + e$
Power function	$y = a + b \cdot x^k + e$
Second-order polynomial function	$y = a \cdot x^2 + b \cdot x + c + e$
Third-order polynomial function	$y = a \cdot x^3 + b \cdot x^2 + c \cdot x + d + e$

Calibration

- The transfer function parameters, e.g., a and b in the linear transfer function, are in general unknown and must be estimated. This process is called **calibration**.
- If a set of known (x,y) paired values are available, estimation of transfer function parameters can be performed by the least square approach.
- For linear transfer function:

$$(\hat{a}, \hat{b}) = \operatorname{argmin}_{a,b} \sum_{i=1}^n [y_i - (a \cdot x_i + b)]^2$$

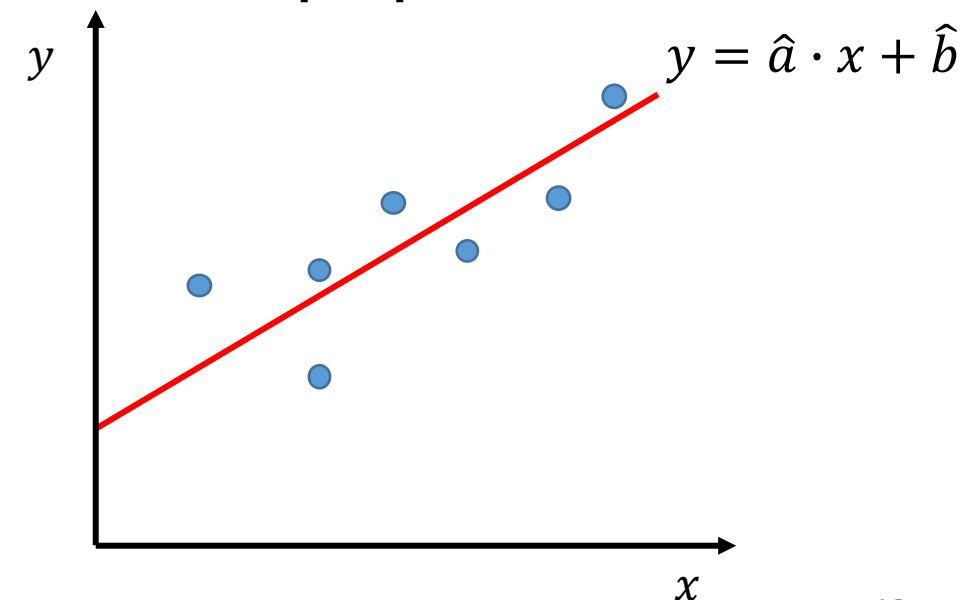
Two-point calibration



$$\hat{a} = \frac{y_2 - y_1}{x_2 - x_1}$$

$$\hat{b} = y_2 - \frac{y_2 - y_1}{x_2 - x_1} \cdot x_2$$

Multiple-point calibration



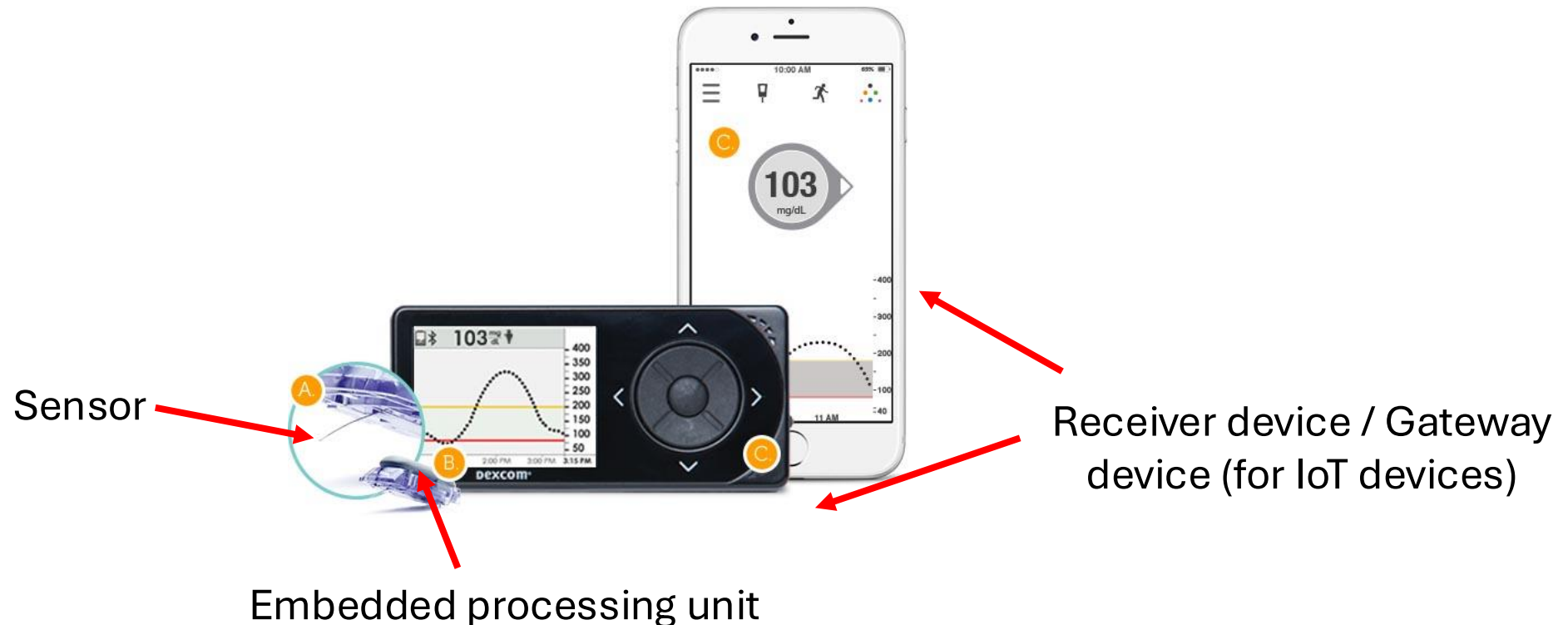
Embedded processing unit

➤ The embedded processing unit is responsible for:

- Performing local processing of the measurements
- Storing some data in a local memory
- Sending the data to another device (e.g., a display device or a gateway device for IoT systems)

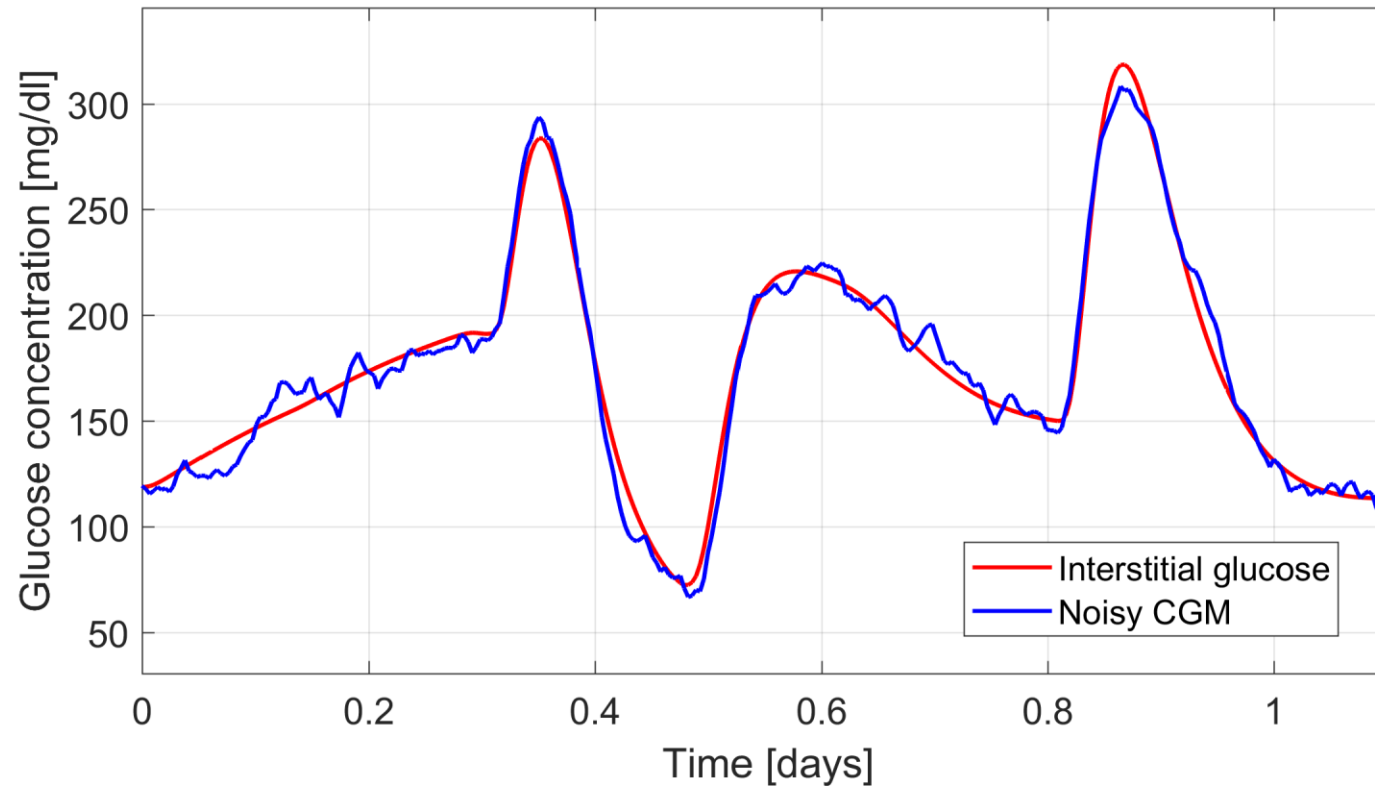
Three main operations:

- Denoising
- Calibration
- Fault detection



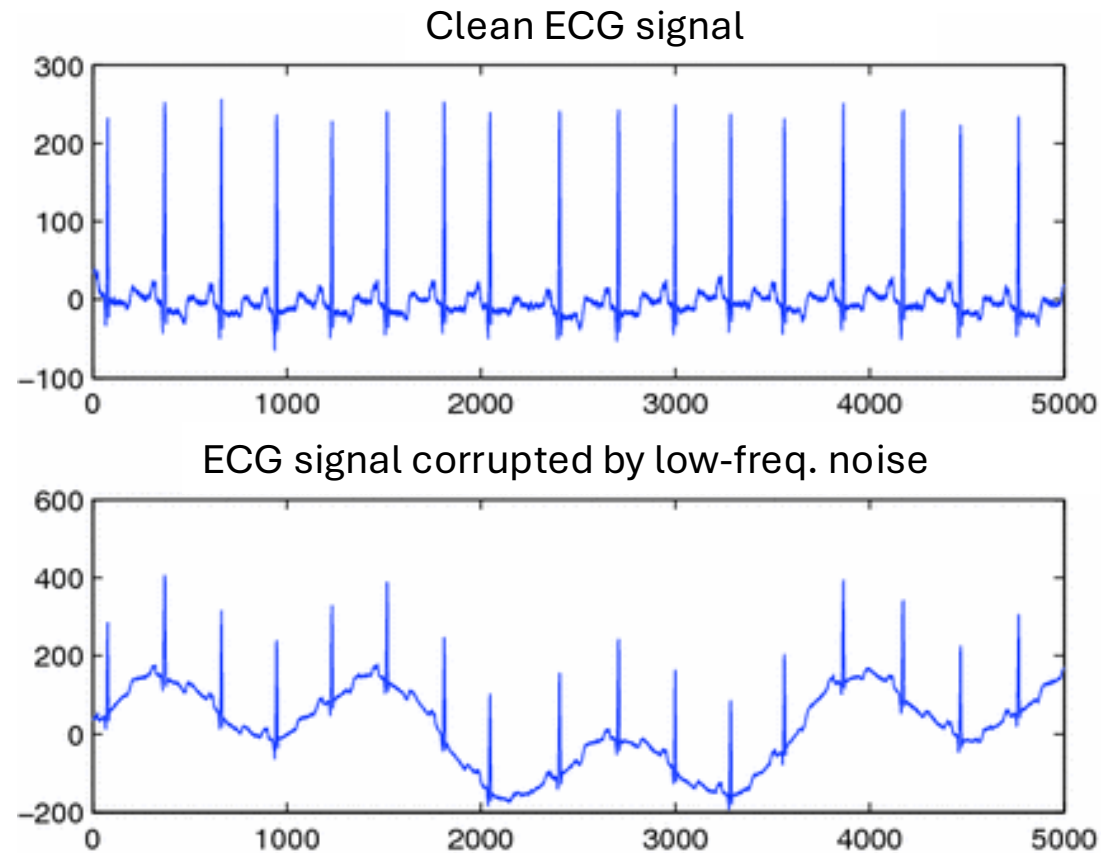
Denoising

- Raw signals are often noisy: the signal of interest is corrupted by noise
- Example: high-frequency noise on a CGM signal



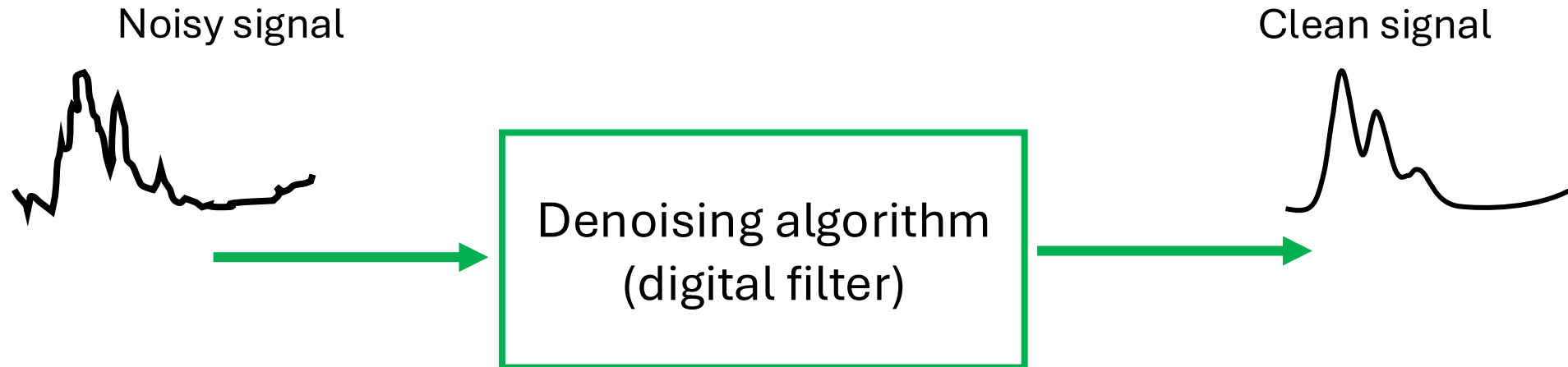
Denoising

- Raw signals are often noisy: the signal of interest is corrupted by noise
- Example: low-frequency noise (e.g., due to movement) on an ECG signal



Denoising

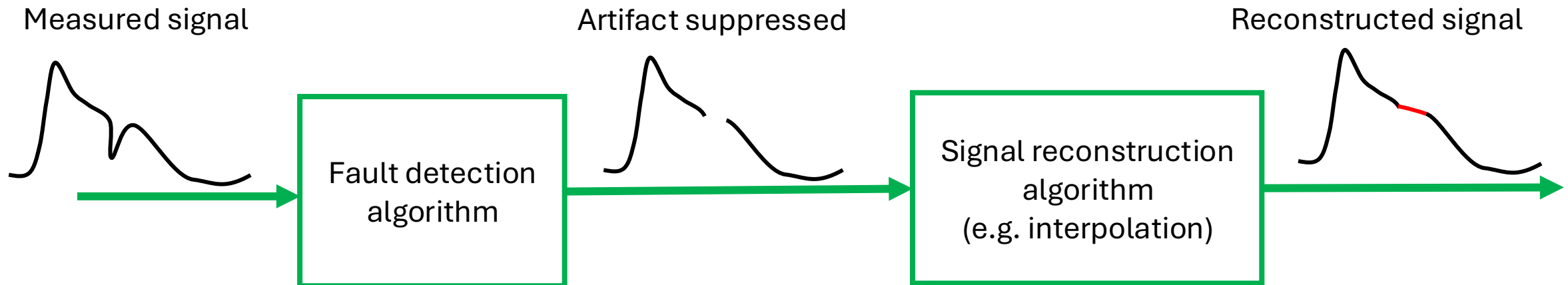
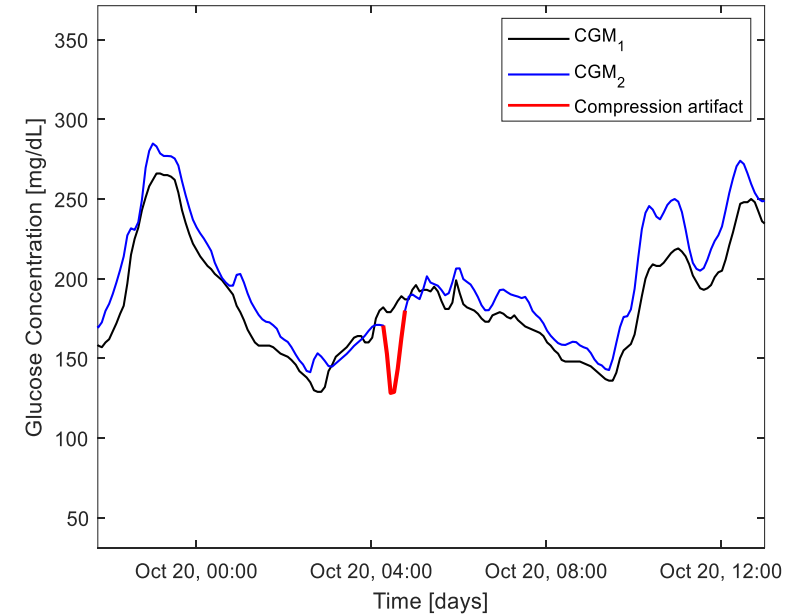
- The noise can be filtered by applying a denoising algorithm exploiting a digital filter.



- Non-causal filters: the filter output depends on past and future inputs → for offline use only
- Causal filters: the filter output depends only on past and present inputs → suitable for online application

Fault detection

- Detection of temporary sensor errors also called faults or artifacts
- Example: pressure-induced signal suppression in a CGM signal

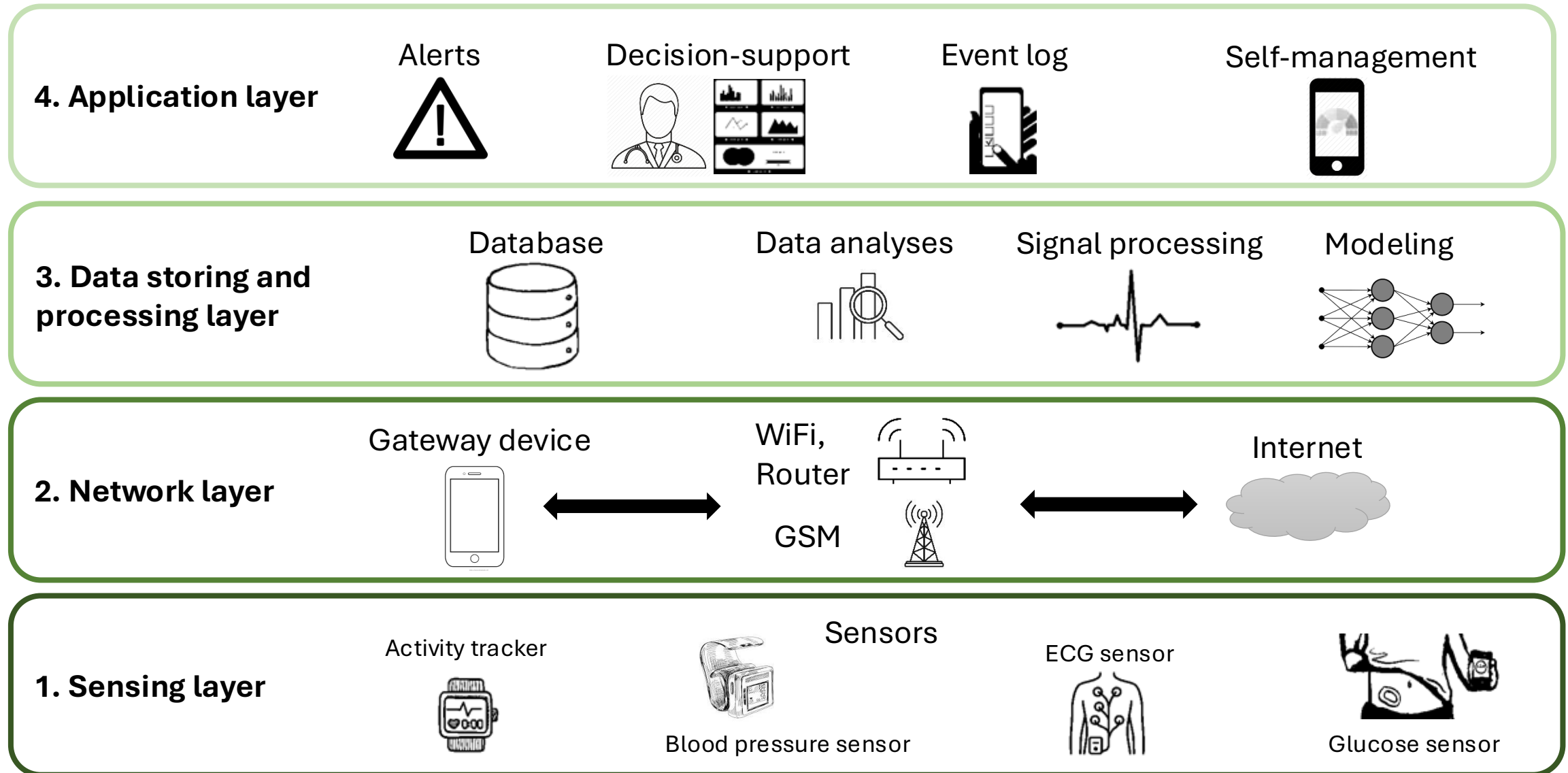


Communication systems

- Wearable devices require a data transmission systems that **wirelessly** communicate between the sensing device and the receiver and/or gateway device (it can be the smartphone).
- **Bluetooth communication** is the primary choice because of its simplicity, low operation cost, compatibility with new smartphone technology, and simple hardware. Drawback: it requires high power consumption.
- **Near field communication (NFC):** a wireless communication technology that operates at a short distance (about 4 cm) and does not need the energy from the battery to transfer the data.
- NFC tags can directly communicate with the smartphone equipped with the NFC modules.
- Bypassing the need for batteries, NFC enables ultrathin, ultraminiaturized designs for sensors, like a temporary transfer tattoo.



The architecture of wearable IoT systems



1. Sensing layer

- This layer includes sensors, which are responsible for:
 - **Collecting the data** either directly from the body through contact sensors or from peripheral sensors providing indirect information of body and behaviours
 - **Preparing the data** for on-board analysis, closed-loop feedback or remote transmission for more comprehensive analysis and storing

- Key components of IoT wearable sensors:
 - Miniaturized **sensor hardware**
 - **Embedded processor** with storage capabilities, power management and communication circuits
- Typically, limited computing power and storage capabilities.

Contact sensors

ECG sensor



Glucose sensor



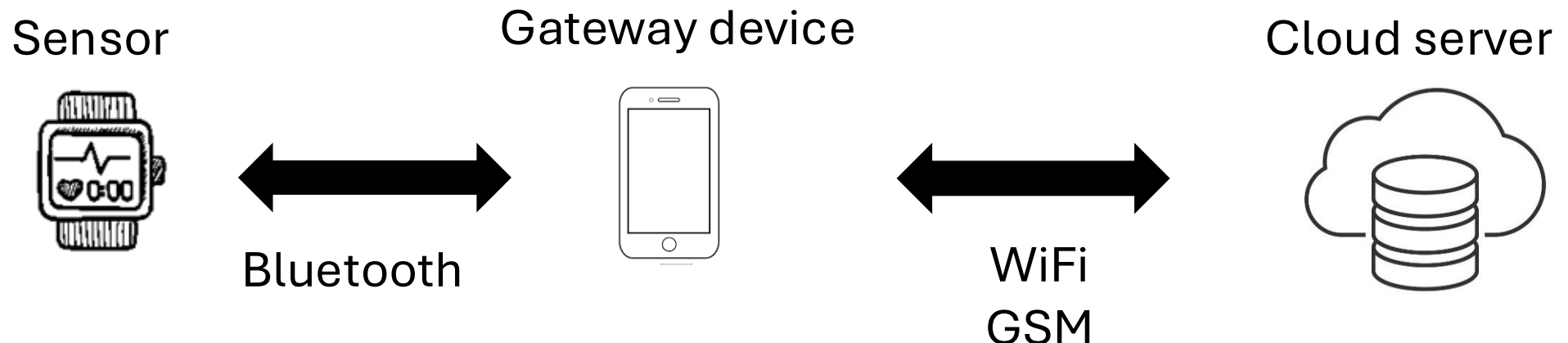
Peripheral sensors

Step counter Microphone and camera



2. Network layer

- For more comprehensive data analysis and storage, the sensor layer needs to transmit the data to **cloud computing servers** (remotely-located computing machines).
- This transmission can be mediated by a companion device that serves as a **gateway**, i.e., it enables the information to flow from sensors to the cloud server.
 - Example of gateway devices: smartphones, tablets and PCs.
- The communication must be **secured** through:
 - Authentication protocols for writing, modifying and accessing the data
 - Security and privacy protocols for preserving the data



3. Data storing and processing layer

- Layer responsible for data storing and complex processing

- Key components:
 - **Database** for structured data storing
 - Data organized based on a specific data model (relational, hierarchical, object-oriented,...)

 - **Algorithms** for data processing
 - Statistical analysis
 - Feature extraction (e.g., heart rate, calories expenditure, ...)
 - Signal processing (e.g., filtering, artifact removal, ...)
 - Data mining (e.g., cluster analysis, anomaly detection, sequential pattern extraction, ...)
 - Data modeling (knowledge-based, data-driven or hybrid approaches)
 - Artificial intelligence (knowledge representation, machine learning, natural language processing, machine perception, automated planning, ...)

3. Data storing and processing layer

➤ Layer responsible for data storing and complex processing

➤ Key components:

Topic not covered

■ **Database** for structured data storing

- Data organized based on a specific data model (relational, hierarchical, object-oriented,...)

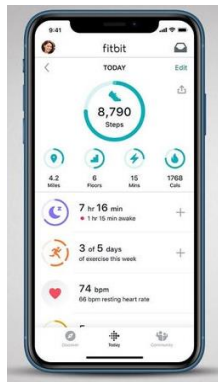
■ **Algorithms** for data processing

- Statistical analysis
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4. Application layer

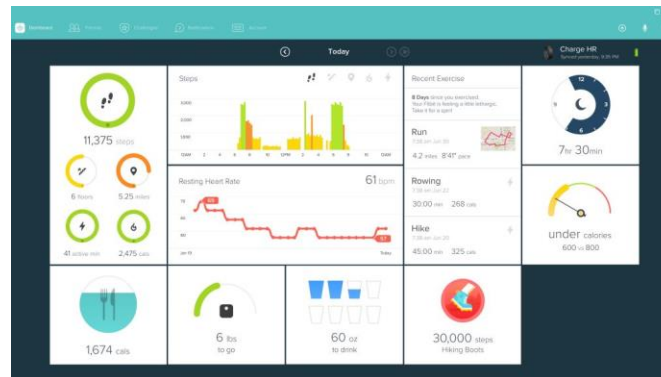
- Layer responsible for visualizing raw or processed data to end-users and for providing decision support to them through a software application
- Possible end-users: patients, physicians, caregivers, analysts, data managers, administrators, etc.
- Possible types of applications:

Mobile app



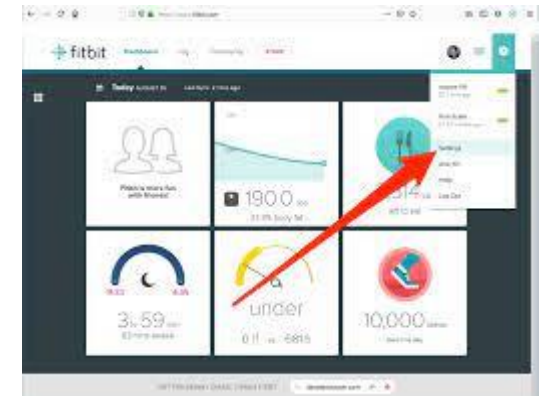
App installed in a mobile device (e.g. smartphone, smartwatch, tablet, etc.)

Desktop app



App installed in a computer. It can work offline.

Web app



App that runs on the browser. Installation not required. Internet access required.